

selfupdate

The Untainted Slate

Status of the runs whose training signal contains NO reference-text labels anywhere: hidden matching and teacher-sourced KL only.

Arms with task-CE readouts (classical distillation-plus-labels in camouflage) are excluded from every conclusion in this document.

Admission rule, per run config:

readout column K (teacher_kl) or — (none); strict / teacher-censored schedules; and all classic_kd runs (teacher-logit KL by definition).

1. The pure slate

admission: no reference-text labels in any training term

LAYERWISE BRANCH — admitted runs (teacher-KL readout, the relabeled KL family):

slide8kl sliding k=8, teacher-KL readout — the deliberate pure arm
k6-KL (slide6pure misfire) sliding k=6, teacher-KL
s43-KL (slide8pure_s43 misfire) seed-43 replication
KL-1.7B (slide8pure_1p7b misfire) the law at 1.7B scale
think-KL (thinkslide misfire) thinking-channel data — premise-condition point
ch8-KL (q_ch8_slide8pure misfire) Quijote prose — the law on a second corpus
thinkselkl thinking-selective data, teacher-KL — see the companion note
pp2-confounded (lw_q_pp2) pipeline-parallel run that silently ran teacher_kl

LAYERWISE BRANCH — admitted controls (no behavioral term at all):

vocab_strict / nmse_strict pure storage, no readout
tc_frozen (teacher_censored) schedule control
lens_kl arms teacher-sourced but UNSAFE (destruction), kept as warnings

CLASSIC_KD BRANCH — all four runs admitted (teacher-logit KL only, by branch doctrine):

kd_full_0p6b · kd 14B LoRA · kd 27B LoRA (early) · kd 30B-A3B LoRA

WHAT THE PURE SLATE ESTABLISHES, WITHOUT ANY LABEL ANYWHERE:

1. STORAGE IS REAL AND LOCATED. Hidden matching writes durable, distributed, redundant weight changes in the upper-middle stack (~80% depth); the classic_kd adapter-norm probes find the same band at 14B-30B. The pure-KL arm's weight profile is bitwise-comparable to the best-reciting arm's (peak L20, 0.0311 vs 0.0310): the knowledge arrives in full.

2. THE LAST-3% LAW. Every pure distribution-matching arm at 0.6B converges to the teacher-fidelity plateau (free-run CER 0.76-0.86 against an untrained 0.897 and a teacher-with-passage ceiling 0.650), with high teacher-forced fidelity (96.5% measured). Seven independent replications across window depth (k=6, k=8), seed (17, 43), scale (1.7B), parallelism (PP2), data channel (thinking) and corpus/genre (Quijote prose, mildly softer at 0.758-0.779).

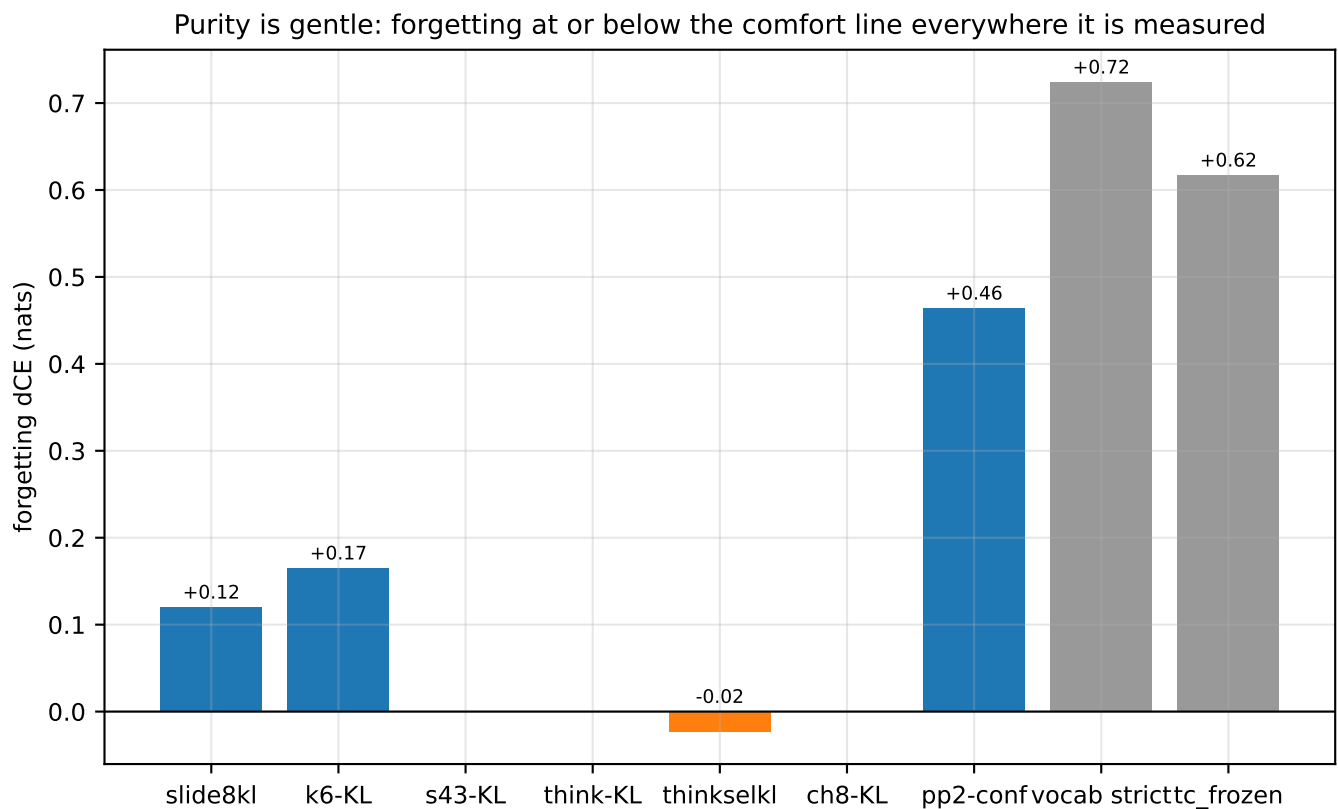
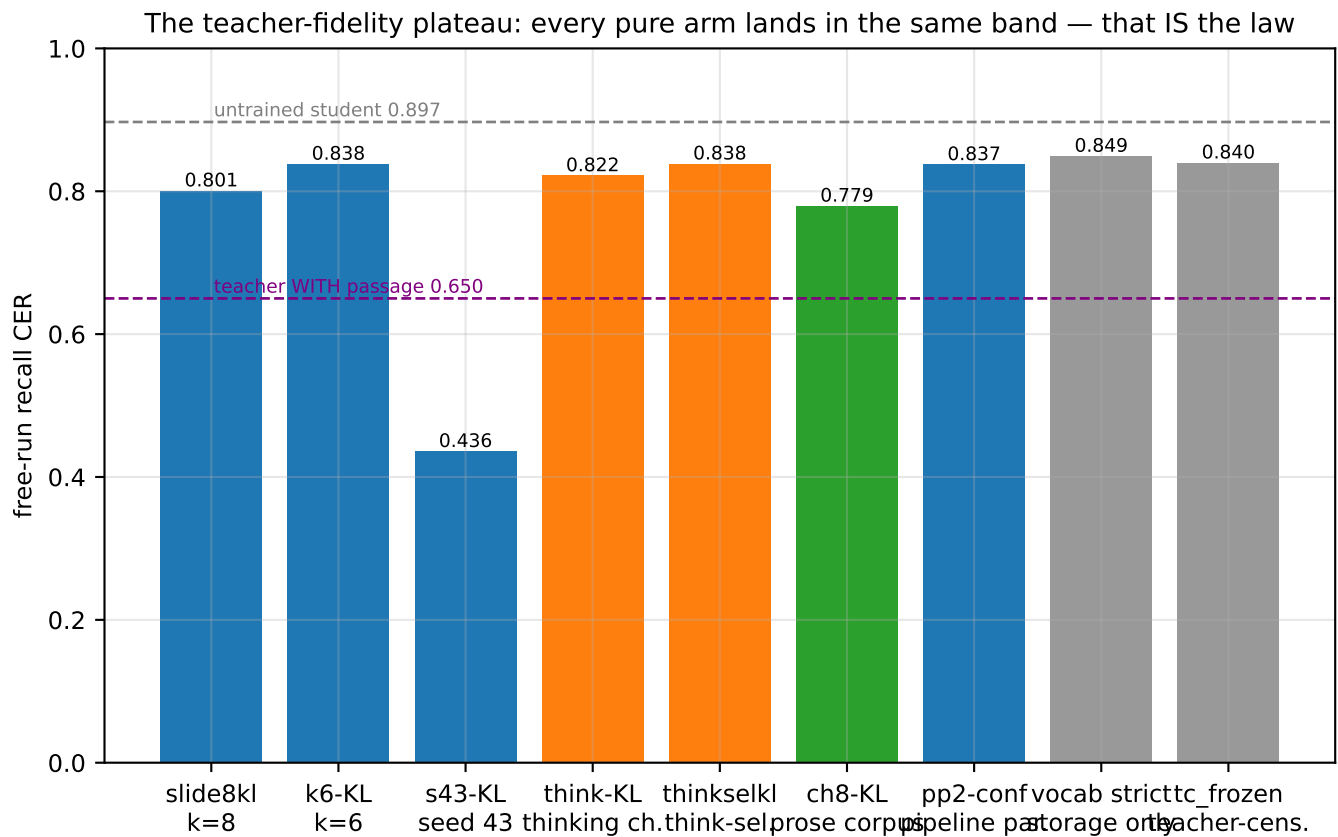
3. THE RESIDUAL CLOSES WITH SCALE. classic_kd, same pure objective, growing student: 0.6B stalls at 0.64; 14B reaches CER 0.016; 30B-A3B reaches 0.000 (line-exact 1.0). No labels were needed - scale (or MoE params-per-item, per the dilution law) buys the last 3% honestly.

4. PURITY IS GENTLE. The pure arms sit at the bottom of the forgetting table (thinkselkl even measured a negative drift, -0.02: nothing behavioral was installed, nothing general was displaced). Cleanliness under the destruction battery is trivial for the plateau arms and near-clean for slide8kl (worst category +0.15).

5. SAFETY IS A LOSS PROPERTY. Within teacher-sourced losses, the distribution-SHAPED ones (fisher-metric, per-layer lens-KL) amplify the intrusion groove (57.5% / 90% intrusion); the geometry-shaped ones (vocab_mse, nmse) are safe. Purity of provenance does not imply safety of shape.

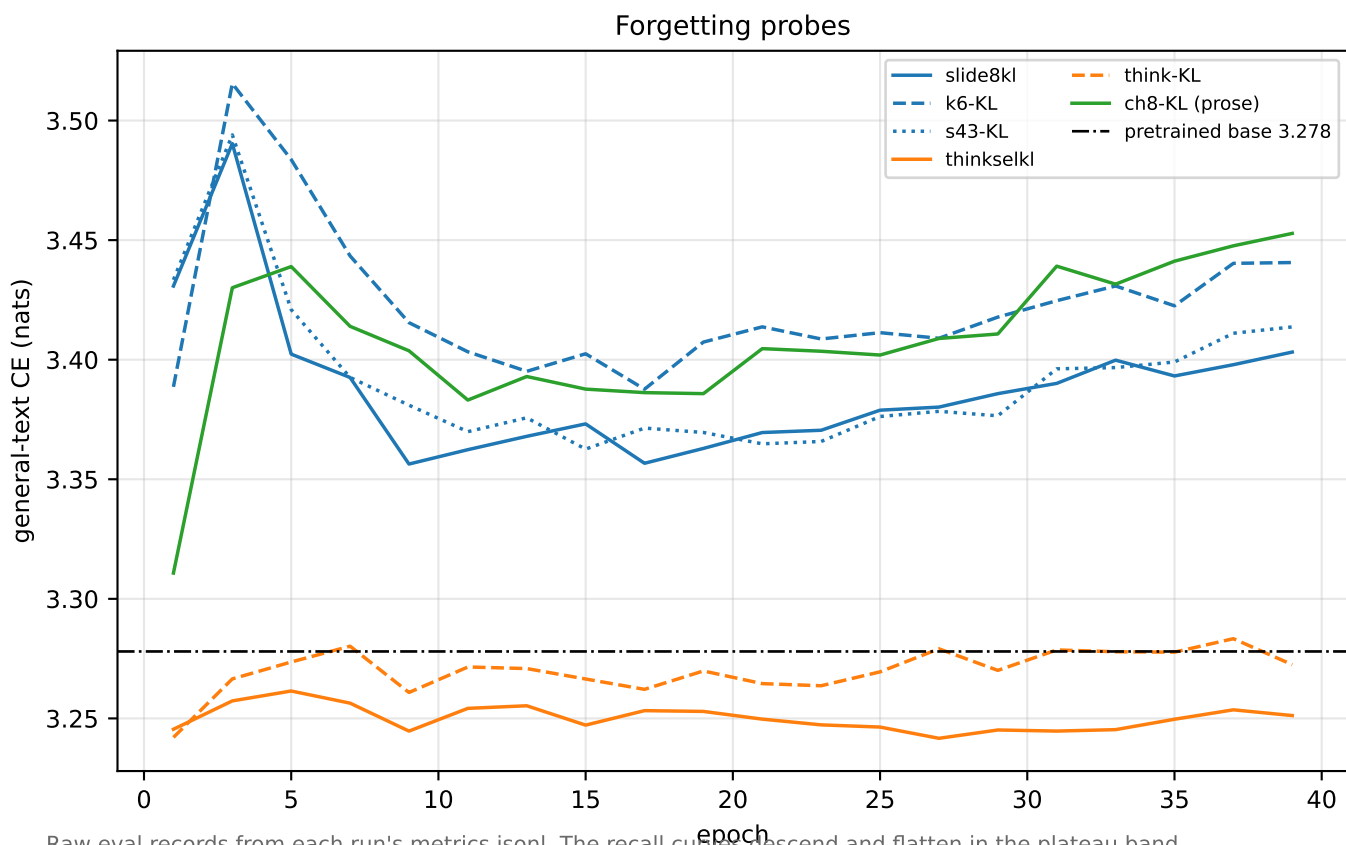
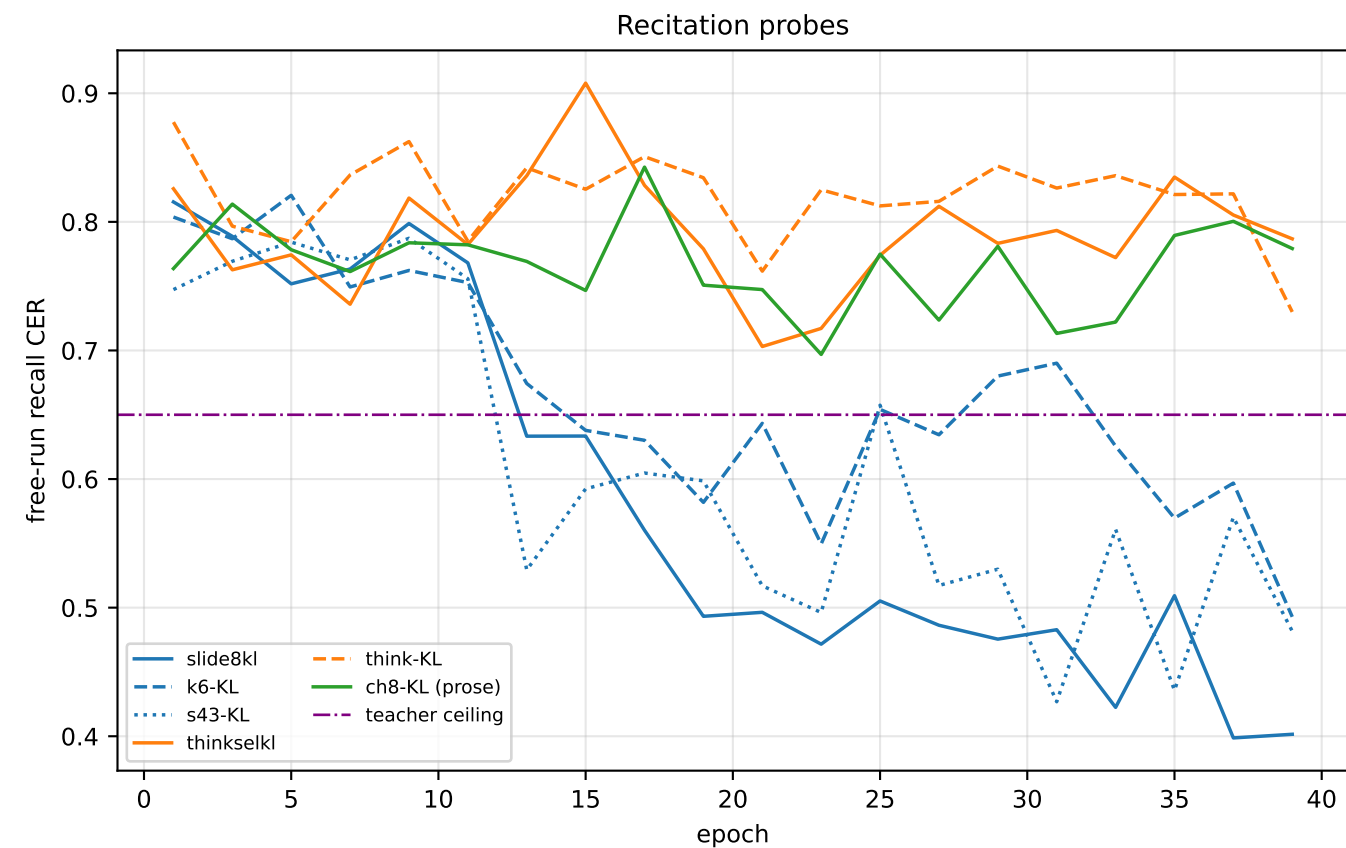
OPEN, ON THE PURE SLATE'S OWN TERMS: (a) the premise-gated thinking teacher (C2-32 fix) - the one pure arm that failed for a diagnosed, repairable reason; (b) layerwise + teacher-KL at 14B-30B, where classic_kd predicts the plateau dissolves - the decisive Campaign 3 experiment; (c) pp2fix certification (parallel training repro, verdict landing unattended).

2. The pure layerwise slate at 0.6B — recall and forgetting



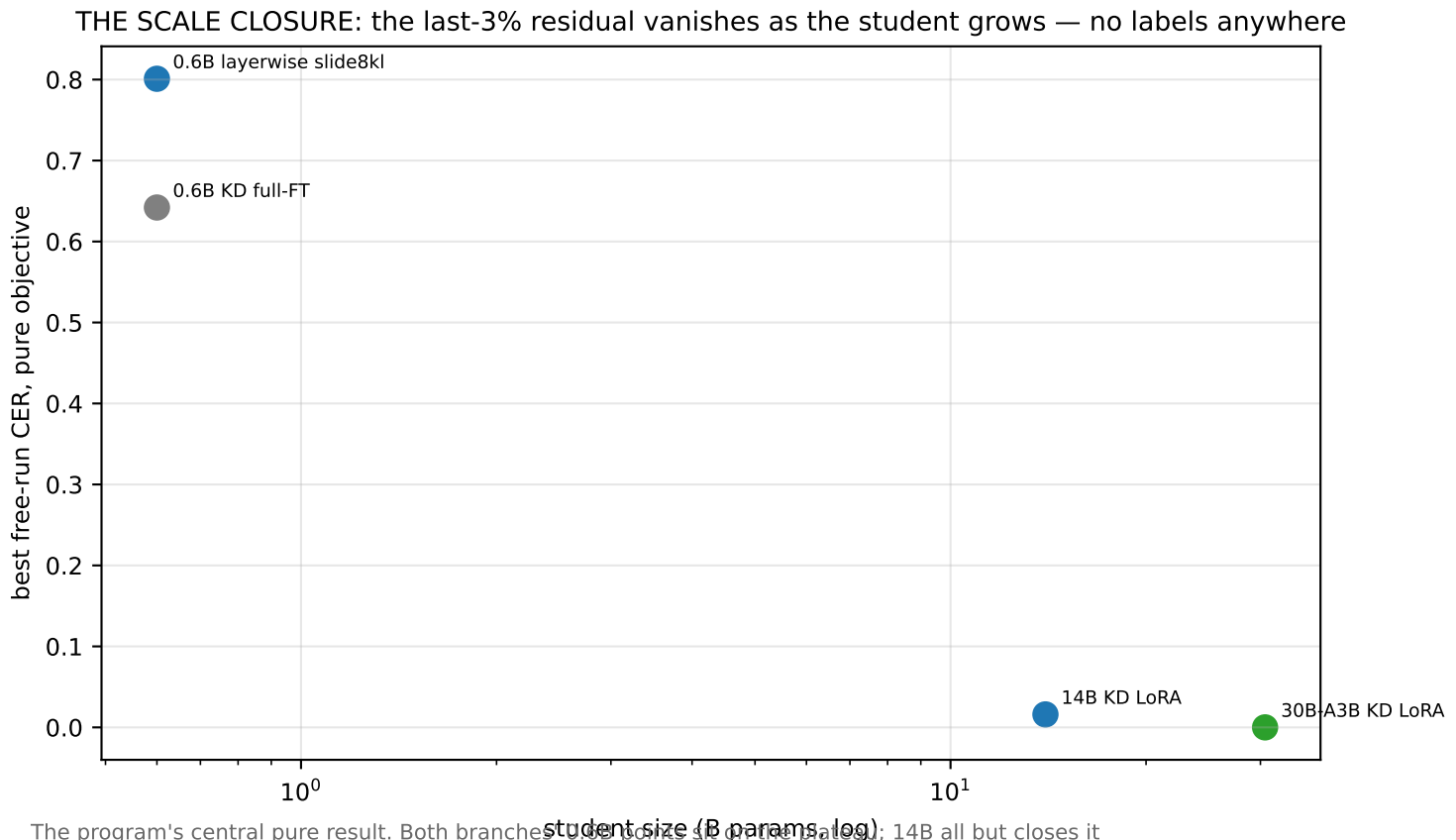
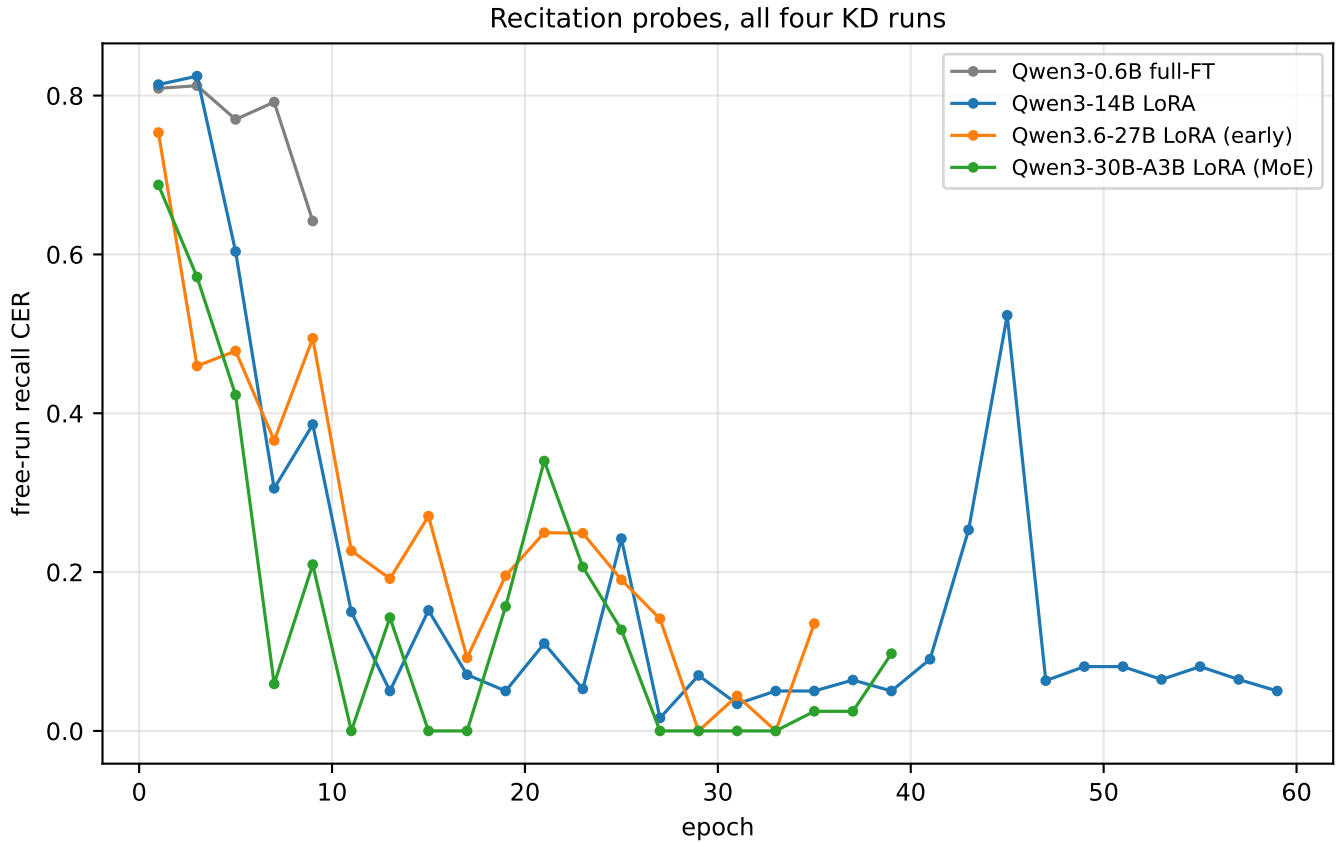
Free-run CER and general-text forgetting for every admitted layerwise arm at 0.6B, from runs/resets.md 9 (full-corpus eval where present, last training eval otherwise). Blue = RAG channel, orange = thinking channel (the premise-condition failures, see companion note), green = Quijote prose corpus, gray = no-readout controls. The plateau band 0.76-0.86 across window depth, seed, parallelism, data channel and corpus is the last-3% law's replication set.

3. Pure-arm dynamics — convergence to the plateau, gentle forgetting



Raw eval records from each run's metrics.jsonl. The recall curves descend and flatten in the plateau band rather than collapsing to zero — smooth, converged training, not failure. The forgetting curves stay near the pretrained baseline: with no label term there is no groove to carve and little general capability displaced.

4. classic_kd — the same pure objective, growing student



The program's central pure result. Both branches close the slide8kl pure battery; 14B all but closes it (0.016, line-exact 0.969); the 30B-A3B MoE closes it exactly (0.000, line-exact 1.000, general-NLL drift +0.22). Owed before celebrating: the destruction/intrusion battery has not yet run on the KD champions — Campaign 2 showed headline recall can hide intrusion.

5. What to run next (pure slate's own agenda)

every item label-free by construction

1. PREMISE-GATED THINKING TEACHER (C2-32 fix; layerwise C3 item 2). thinkselkl failed for a diagnosed mechanical reason - the thinking teacher's context lacked the passage, so its distribution at answer positions was diffuse and KL had nothing to transmit. Carry the passage in the thinking teacher sequence (or premise-gate per item) and rerun. This is the cheapest possible repair of a pure arm, and the thinking channel is the gentlest data channel measured (see companion note).
2. LAYERWISE + TEACHER-KL AT 14B-30B (the decisive experiment). classic_kd shows the plateau dissolving with scale under whole-network KL. If layerwise hidden matching + teacher-KL readout rides the same closure at 14B+, the program has a label-free, bounded-backward-depth recipe - exactly what the 120B Quijote stage needs. The Qwen3.6-27B bridge grid is already queued.
3. BATTERY ON THE KD CHAMPIONS. CER 0.000 at 30B is unaudited for damage: run the category-resolved destruction battery and the n=200 intrusion bait on the 30B-A3B and 14B checkpoints.
4. ANCHOR-KL ON A PURE ARM. The anchor-KL regularizer is teacher-sourced (KL to the base model's own distribution on anchor texts) and was only ever measured on hybrid carriers. Diverse anchors at w=1.0 cost nothing and halved damage on the hybrid - measure it on slide8kl / the 14B KD recipe (see companion note).
5. PP2FIX CERTIFICATION verdict (landing unattended) - unlocks parallel training for the scale runs.
6. TWO-PHASE, PURIFIED. The 2p separability result (body first, readout after, on a frozen body) was demonstrated with a label readout. Rerun phase 2 as teacher-KL: if post-hoc readout installation works label-free at 14B+, storage and readout can be SCHEDULED independently at the Quijote stage (stream blocks for storage, one short readout pass at the end).